Venn Diagrams of three sets

The **intersection** of three sets X, Y and Z is the set of elements that are common to sets X, Y and Z. It is denoted by $X \cap Y \cap Z$

Example:

Draw a Venn diagram to represent the relationship between the sets

$X = \{1, 2, 5, 6, 7, 9\}$, $Y = \{1, 3, 4, 5, 6, 8\}$ and

$Z = \{3, 5, 6, 7, 8, 10\}$

Solution:

We find that $X \cap Y \cap Z = \{5, 6\}$, $X \cap Y = \{1, 5, 6\}$,

$Y \cap Z = \{3, 5, 6, 8\}$ and $X \cap Z = \{5, 6, 7\}$

For the Venn diagram:

Step 1: Draw three overlapping circles to represent the three sets.

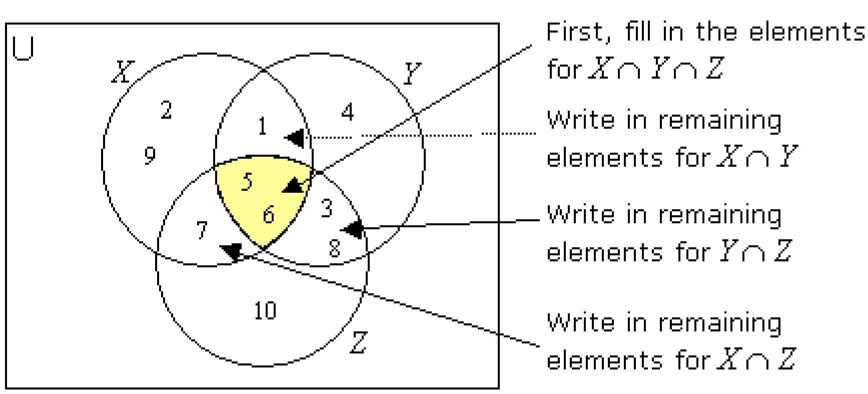
Step 2: Write down the elements in the intersection ($X \cap Y \cap Z$)

Step 3: Write down the remaining elements in the intersections:

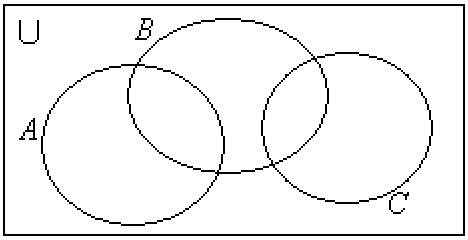
($X \cap Y$, $Y \cap Z$ and $X \cap Z$

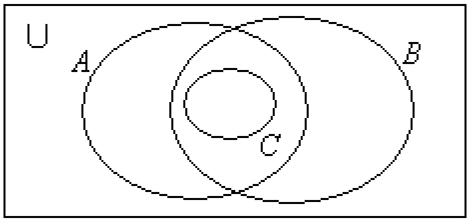
Step 4: Write down the remaining elements in the respective sets.

Again, notice that you start filling the Venn diagram from the elements in the intersection first.



In general, there are many ways that 3 sets may intersect. Some examples are shown below.





How to shade regions of Venn Diagrams involving three sets

Shading Regions with Three Sets,

Example:
Shade the indicated region:
1) (A ∩ B) ∩ C
2) (A ∪ B) ∩ C

Shading Regions with Three Sets,

Example:
Shade the indicated region:
3) (A ∪ B)' ∩ C
4) (A' ∩ B') ∩ C'

How to solve word problems with 3-Set Venn Diagrams?

Venn Diagram Problem with 3 Circles
Use the given information to fill in the number of elements in each region of the Venn Diagram.

Example 1:
150 college freshmen were interviewed. 85 were registered for a math class, 70 were registered for an English class and 50 were registered for both math and English

a) How many signed up only for a math class?
b) How many signed up only for an English class?
c) How many signed up for math or English?
d) How many signed up for neither math nor English?

Example 2:
100 were students interviewed. 28 took PE, 31 took Bio, 42 took Eng, 9 took PE and Bio, 10 took PE and Eng, 6 took Bio and Eng and 4 took all three subjects.

a) How many students took none of the three subjects?
b) How many students took PE, but not Bio or Eng?
c) How many students took Gio and PE but not Eng?

Example 3:
110 college freshmen were surveyed. 25 took physics, 45 took biology, 45 took mathematics, 10 took physics and mathematics, 8 took biology and mathematics, 6 took physics and biology and 5 took all three.

a) How many students took biology, but neither physics nor mathematics?
b) How many students took biology, physics or mathematics?
c) How many students did not take any of the three subjects?